

EXPERIMENT NO. 7 — APRIORI ALGORITHM FOR MARKET BASKET ANALYSIS

Learning Objective: Student should be able to implement Apriori Algorithm for Market Basket Analysis.

Tools: C / C++ / Python / Data Mining Tool

Theory: Implementation of Apriori Algorithm for Market Basket Analysis.

1. Apriori Algorithm

Apriori is a classic association-rule mining algorithm introduced by R. Agrawal and R. Srikant. It is used to discover **frequent itemsets** and generate association rules from transactional databases. A transaction represents a set of items purchased or occurring together, such as products in a customer's shopping basket.

The main idea of Apriori is to discover frequent itemsets level by level. It first finds frequent 1-itemsets, then uses them to generate candidate 2-itemsets, then candidate 3-itemsets, and continues until no more frequent itemsets can be found. Candidate itemsets whose support is below the minimum support threshold are removed.

The algorithm is called Apriori because it uses prior knowledge about frequent itemsets to reduce the search space. Instead of checking every possible combination blindly, it eliminates candidates that cannot possibly become frequent.

2. Market Basket Analysis

Market Basket Analysis is a data-mining technique used to identify relationships between items that occur together in transactions. For example, a supermarket may discover that customers who buy bread frequently also buy butter. Such relationships can be used for product placement, recommendation systems, promotions, cross-selling and inventory planning.

The input is generally a transaction database in which each transaction contains a set of items. The output consists of frequent itemsets and, when association rules are generated, rules such as **{Bread} → {Butter}**.

3. Important Terms

Term	Meaning
Item	A single product or element, e.g. Milk.
Itemset	A set of one or more items, e.g. {Milk, Bread}.
Transaction	A collection of items purchased/observed together.
Support	Frequency with which an itemset occurs in the transaction database.
Confidence	Strength of an association rule, i.e. how often the consequent occurs when the antecedent occurs.
Lift	Measures how much more often X and Y occur together than expected if they were independent.
Minimum Support	Threshold used to decide whether an itemset is frequent.
Minimum Confidence	Threshold used to decide whether an association rule is strong enough.

4. Apriori Property

The most important property of Apriori is the **downward-closure / anti-monotone property**: **every non-empty subset of a frequent itemset must also be frequent.**

Equivalently, if an itemset is not frequent, then none of its supersets can be frequent. This property allows Apriori to prune a large number of candidates without counting their support.

If {A, B, C} is frequent,
then {A,B}, {A,C}, {B,C},
{A}, {B}, and {C} must also be frequent.

If {A,B} is NOT frequent,
then {A,B,C} cannot be frequent.

This pruning rule is the main reason Apriori is practical compared with checking every possible item combination.

APRIORI — THEORY AND ALGORITHM

5. Support

Support indicates how frequently an itemset appears in the complete transaction database.

$$\text{Support}(X) = \frac{\text{Number of transactions containing } X}{\text{Total number of transactions}}$$

$$\text{Support percentage} = \text{Support}(X) \times 100$$

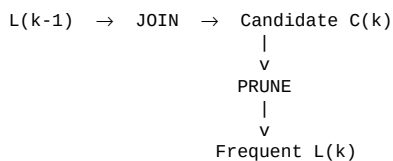
For example, if {Milk, Bread} occurs in 4 out of 10 transactions, its support is $4/10 = 0.40$ or 40%. If minimum support is 30%, the itemset is frequent.

6. Candidate Generation — Join and Prune

Apriori uses two major operations at each level: **join** and **prune**.

Join step: frequent $(k-1)$ -itemsets are combined to form candidate k -itemsets.

Prune step: a candidate is removed if any of its $(k-1)$ -itemset subsets is not present in the previous frequent-itemset set.



7. Apriori Algorithm Steps

- Step 1: Scan the transaction database.
- Step 2: Generate C_1 , the candidate 1-itemsets.
- Step 3: Count support of every candidate.
- Step 4: Remove candidates below minimum support to obtain L_1 .
- Step 5: Join frequent $(k-1)$ -itemsets to create C_k .
- Step 6: Prune C_k using the Apriori property.
- Step 7: Count support of remaining candidates.
- Step 8: Remove candidates below minimum support to obtain L_k .
- Step 9: Repeat until no frequent itemsets remain.

Each pass over the transaction database produces the frequent itemsets of the current size. The process terminates when the generated candidate set becomes empty or no candidate satisfies minimum support.

8. Association Rules

After frequent itemsets are discovered, they can be used to generate association rules. A rule has the form $X \rightarrow Y$, where X and Y are disjoint itemsets.

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$$
$$\text{Lift}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X) \times \text{Support}(Y)}$$

A confidence value of 80% means that among transactions containing X , 80% also contain Y . Lift greater than 1 indicates a positive association relative to independence; lift equal to 1 suggests independence; and lift below 1 indicates a negative association.

9. Advantages

- Simple and easy to understand because it works level by level.
- Uses the Apriori property to eliminate impossible candidates.
- Useful for discovering frequent combinations in transactional data.

- Can produce interpretable association patterns.
- Widely applicable to retail, recommendation and clickstream analysis.

10. Limitations

- Repeated scans of the database can be expensive.
- A large number of candidate itemsets may be generated.
- Performance can degrade when the minimum support threshold is very low.
- Large transaction databases may require considerable memory and computation.
- Other algorithms such as FP-Growth can reduce candidate-generation overhead.

11. Applications

- Retail market basket analysis.
- Product recommendation systems.
- Cross-selling and promotional planning.
- Website click-pattern analysis.
- Medical and biological association analysis.
- Fraud and event-pattern discovery.

PART B — SAMPLE EXAMPLE

Consider the following transaction database. Each transaction contains products purchased together. Let the **minimum support count = 2**. Therefore, an itemset must occur in at least two transactions to be considered frequent.

Transaction	Items
T1	I1, I2, I5
T2	I2, I4
T3	I2, I3
T4	I1, I2, I4
T5	I1, I3
T6	I2, I3
T7	I1, I3
T8	I1, I2, I3, I5
T9	I1, I2, I3

12. Step 1 — Candidate 1-Itemsets (C1)

Count each individual item:

Item	Support Count	Frequent?
I1	6	Yes
I2	7	Yes
I3	6	Yes
I4	2	Yes
I5	2	Yes

Since the minimum support count is 2, all five items survive. Therefore:

$L_1 = \{I1, I2, I3, I4, I5\}$

13. Step 2 — Candidate 2-Itemsets (C2)

All possible pairs are generated from L_1 . Their support counts are obtained by scanning the transactions.

2-Itemset	Support Count	Frequent?
{I1,I2}	4	Yes
{I1,I3}	4	Yes
{I1,I4}	1	No
{I1,I5}	2	Yes
{I2,I3}	4	Yes
{I2,I4}	2	Yes
{I2,I5}	2	Yes
{I3,I4}	0	No
{I3,I5}	1	No
{I4,I5}	0	No

Therefore:

$L_2 = \{$
 {I1, I2},
 {I1, I3},
 {I1, I5},
 {I2, I3},
 {I2, I4},
 {I2, I5}
}

14. Step 3 — Candidate 3-Itemsets

Candidates are generated from L_2 . Before counting support, Apriori pruning removes a candidate if one of its 2-item subsets is not frequent.

For example, {I1,I2,I4} has subsets {I1,I2}, {I1,I4}, and {I2,I4}. Since {I1,I4} is not frequent, {I1,I2,I4} is pruned immediately.

Similarly, after join-and-prune, the useful candidates include combinations whose every 2-item subset is present in L_2 .

3-Itemset	Support Count	Frequent?
{I1,I2,I3}	2	Yes
{I1,I2,I5}	2	Yes

3-Itemset	Support Count	Frequent?
{I1,I3,I5}	1	No
{I2,I3,I5}	1	No

15. Step 4 — Final Frequent Itemsets

Level	Frequent Itemsets
L1	{I1}, {I2}, {I3}, {I4}, {I5}
L2	{I1,I2}, {I1,I3}, {I1,I5}, {I2,I3}, {I2,I4}, {I2,I5}
L3	{I1,I2,I3}, {I1,I2,I5}

APRIORI — EXAMPLE CONTINUED

16. Support Percentage Example

There are 9 transactions in the database. For the itemset {I1,I2}:

Support count = 4

$\text{Support}(\{I1, I2\})$
 $= 4 / 9$
 ≈ 0.4444

Support percentage
 $\approx 44.44\%$

Since 44.44% is above the minimum support threshold represented by $2/9 \approx 22.22\%$, {I1,I2} is frequent.

17. Example Association Rule

Suppose we form the rule {I1} → {I2}. From the database:

$\text{Support}(I1) = 6/9$
 $\text{Support}(I2) = 7/9$
 $\text{Support}(I1, I2) = 4/9$

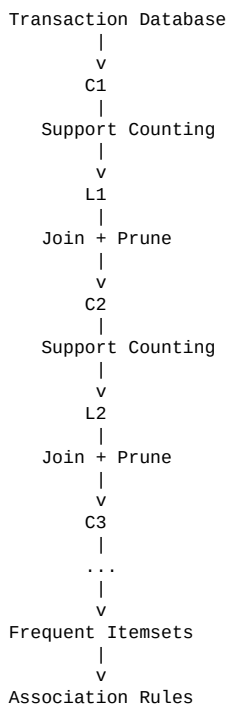
$\text{Confidence}(I1 \rightarrow I2)$
 $= \text{Support}(I1, I2) / \text{Support}(I1)$
 $= (4/9) / (6/9)$
 $= 4/6$
 $\approx 66.67\%$

The rule therefore has a confidence of approximately 66.67%. Its lift is:

$\text{Lift}(I1 \rightarrow I2)$
 $= (4/9) / [(6/9) \times (7/9)]$
 ≈ 0.8571

A lift below 1 in this particular example indicates that the co-occurrence is lower than would be expected if the two items were independent. This illustrates why confidence alone is not sufficient to judge the usefulness of a rule.

18. Apriori Process Summary



19. Result and Conclusion

The Apriori algorithm was studied and applied to a transactional database for Market Basket Analysis. Frequent itemsets were generated level by level using minimum support count, while the Apriori property was used to prune candidates whose subsets were not frequent.

The experiment demonstrates how frequently co-occurring products can be discovered from transaction data. The resulting frequent itemsets can further be used to generate association rules using measures such as support, confidence and lift.